Fields and Waves I

Lecture 14

Current & Resistance

Alex Patterson
Electrical, Computer, and Systems Engineering Department
Rensselaer Polytechnic Institute, Troy, NY

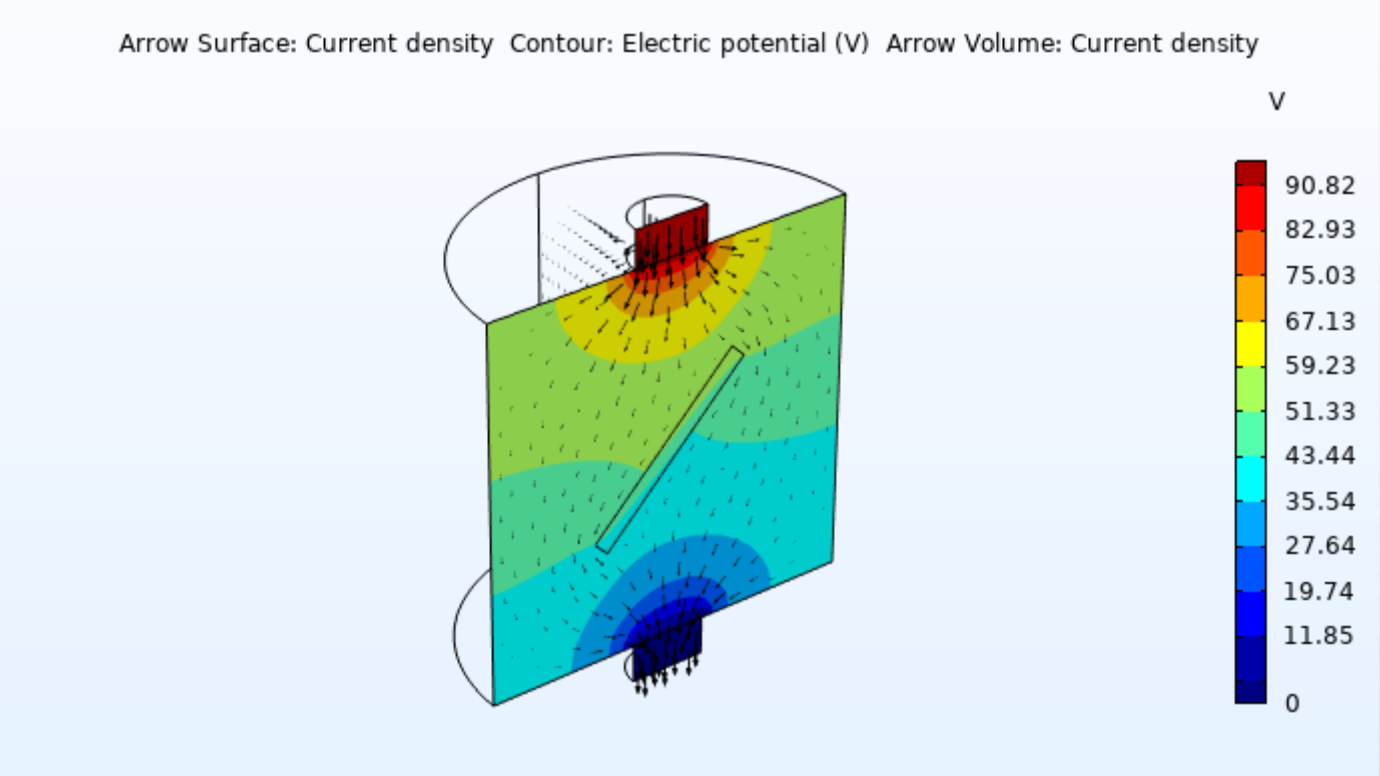
These slides were prepared through the work of the following people:

- Kenneth A. Connor – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- J. Darryl Michael – GE Global Research Center, Niskayuna, NY
- Thomas P. Crowley – National Institute of Standards and Technology, Boulder, CO
- Sheppard J. Salon – ECSE Department, Rensselaer Polytechnic Institute, Troy, NY (Emeritus)
- Lale Ergene – ITU Informatics Institute, Istanbul, Turkey
- Jeffrey Braunstein – ECE Department, University at Albany
- James Lu - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- James Dylan Rees - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY
- Alex Patterson - ECSE Department, Rensselaer Polytechnic Institute, Troy, NY

Materials from other sources are referenced where they are used.
Those listed as Ulaby are figures from Ulaby's textbook.

Overview

- Current
- Resistance



https://www.comsol.com/support/learning-center/course/introduction-to-modeling-resistive-and-capacitive-devices-132/introduction-to-modeling-resistive-and-capacitive-devices-27591

Laplace and Poisson Equations

Review

- We can directly calculate the potential from the charge density ρ using:

- Laplace's Equation:

$$\nabla^2 V = 0$$

- Poisson's Equation

$$\nabla^2 V = -\frac{\rho}{\epsilon}$$

Laplace and Poisson Equations

Review - Boundary Conditions

- Dielectric-Dielectric

	Normal To Boundary	Tangential To Boundary
Fields	$D_{1n} = D_{2n}$	$E_{1t} = E_{2t}$
Potential	$\epsilon_1 \frac{\partial V_1}{\partial n} = \epsilon_2 \frac{\partial V_2}{\partial n}$	$V_1 = V_2$

Laplace and Poisson Equations

Review - Boundary Conditions

- Dielectric-Conductor

	Normal To Boundary	Tangential To Boundary
Fields	$D_{1n} = \rho_s$	$E_{1t} = 0$
Potential	$\epsilon_1 \frac{\partial V_1}{\partial n} = \rho_s$	$V_1 = constant$

Current and Resistance

Current and Resistance

- We have been considering electrostatics - the case in which there are no moving charges.
- This obviously doesn't hold in general. When charges move, we have current.
- We will consider two parameters:
  - Current I (measured in amperes)
  - Current Density $\vec{J}$ (measured in amperes per unit area)

Current and Resistance

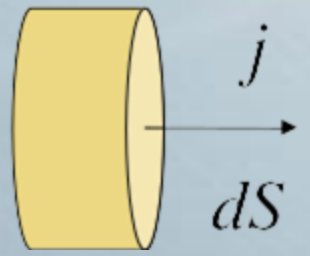
- Current (I) is the flux integral of current density ($\vec{J}$) [Amperes/area]:

$$I = \int j \bullet ds$$

Example: wire with constant current density $j = j_0 \cdot \hat{a}_z$

$$I = \int_0^{2\pi} \int_0^{a} j_0 \cdot \hat{a}_z \bullet r \cdot dr \cdot d\varphi \cdot \hat{a}_z = j_0 \cdot \pi \cdot a^2$$

ds , in cylindrical geometry



Current and Resistance

$j = \sigma \cdot E$ = Ohm's Law

Conductivity - units of S/m or 1/ohm-m

- varies from 10^{7} to 10^{-15}

Good conductor eg. Cu

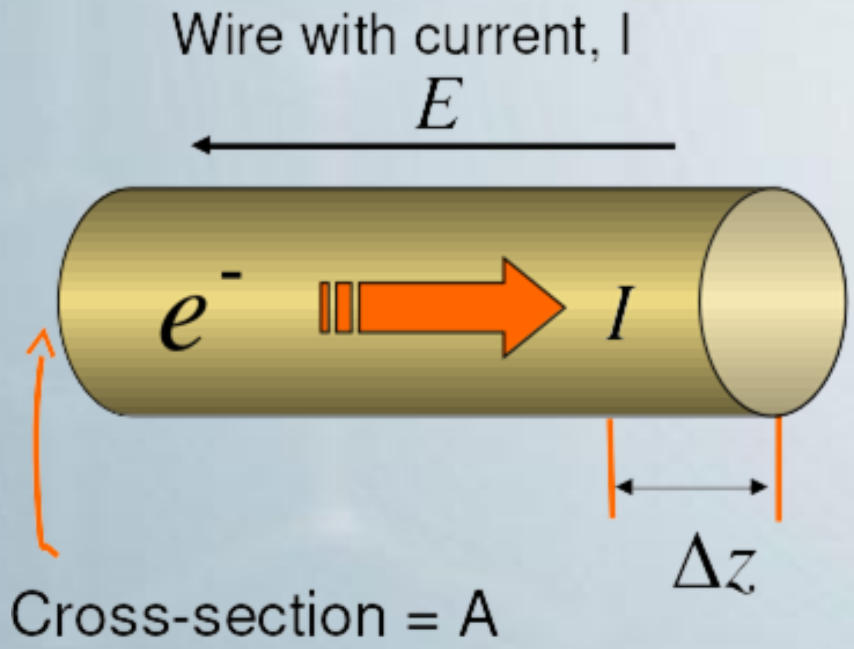
Good insulator

$j = \sigma \cdot E$, is Fields and Waves version of Ohm's Law

$A/m^2 = 1/\Omega m \sqrt{1/m}$

Current and Resistance

- Where does the relationship $\vec{j} = \sigma \vec{E}$ come from? What is σ physically?



$$I = \frac{\Delta Q}{\Delta t} \Rightarrow j = \frac{\Delta Q}{A \cdot \Delta t}$$

Assume all particles move at same *v*

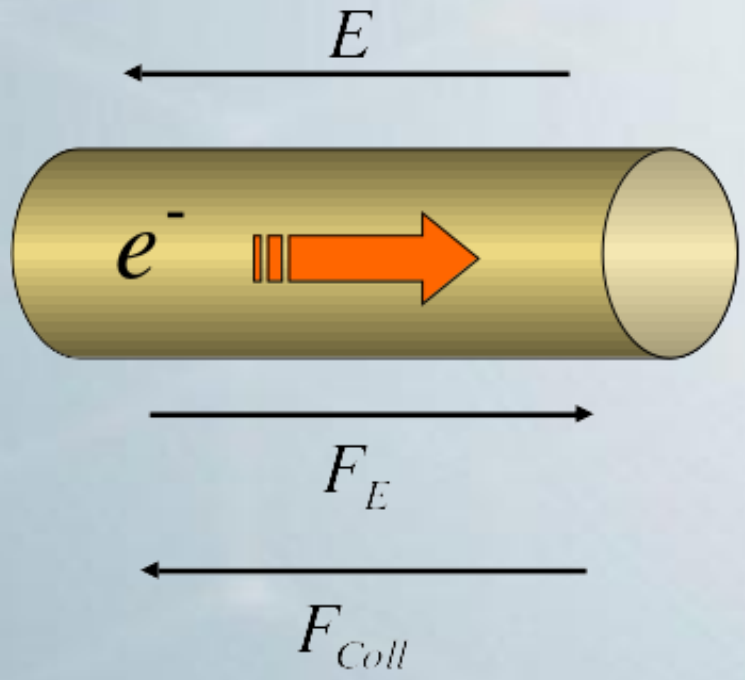
- In Δt, all particles within Δz = v.Δt will pass through the right face

$$\therefore \Delta Q = \rho \cdot \Delta z \cdot A = \rho \cdot v \cdot A \cdot \Delta t$$

$$\Rightarrow j = \frac{\Delta Q}{A \cdot \Delta t} = \frac{\rho \cdot v \cdot A \cdot \Delta t}{A \cdot \Delta t} = \rho \cdot v \Rightarrow j = \rho \cdot v$$

Can we define this speed ?

Current and Resistance



E , in wire pushed electrons to the right

F_E , force (from E-field) driving electrons

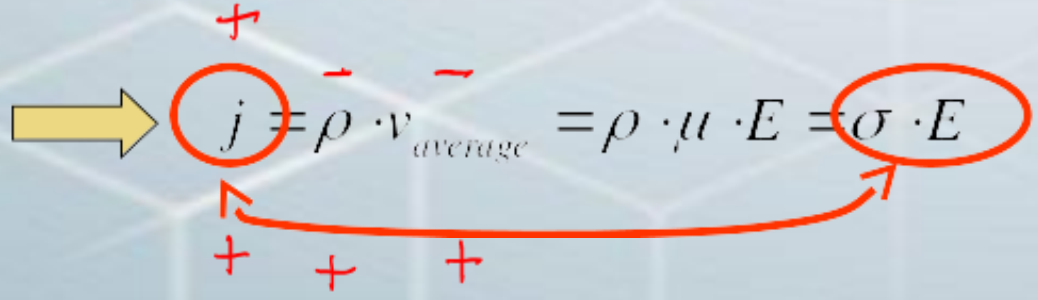
e^- s, collide with lattice ions

F_{Coll} , balances F_E at some $v_{average}$

$$v_{average} = \mu \cdot E$$

(μ= electron mobility, not permeability)

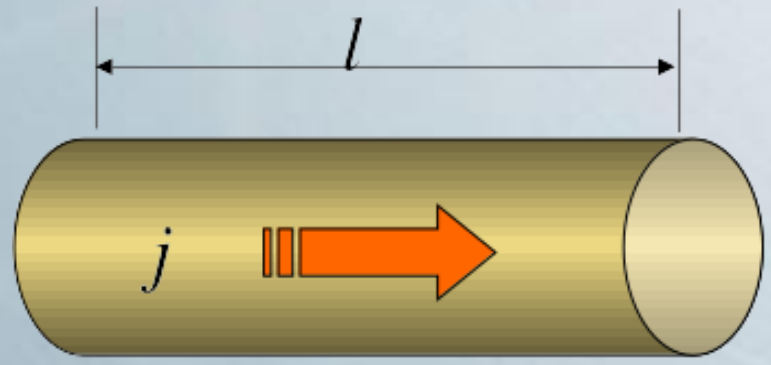
Since $j = \rho \cdot v_{average}$



Current and Resistance

- How can we calculate the resistance of a cylindrical wire with constant current density?

$$R = \frac{V}{I}$$



$\vec{J} = \sigma \vec{E}$ constant

$S/m \cdot V/m \cdot m^2 = S \cdot V = A$

V/Ω

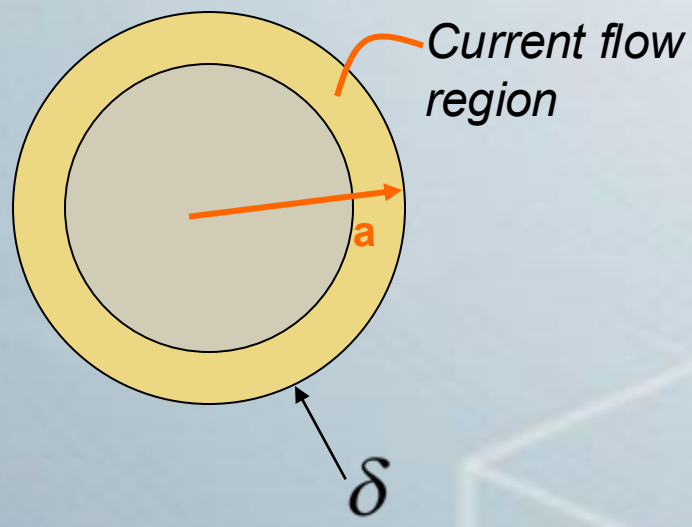
$$I = j \cdot A = \sigma \cdot E \cdot A$$

$$V = -\int E \bullet dl = E \cdot l$$

$$\frac{V}{I} = \frac{E \cdot l}{\sigma \cdot E \cdot A} = \frac{l}{\sigma \cdot A} = R$$

Current and Resistance

- However, current density is rarely constant.
- For example, in conductors, current tends to flow in increasingly thin surface sheets at higher frequencies (the "skin effect")

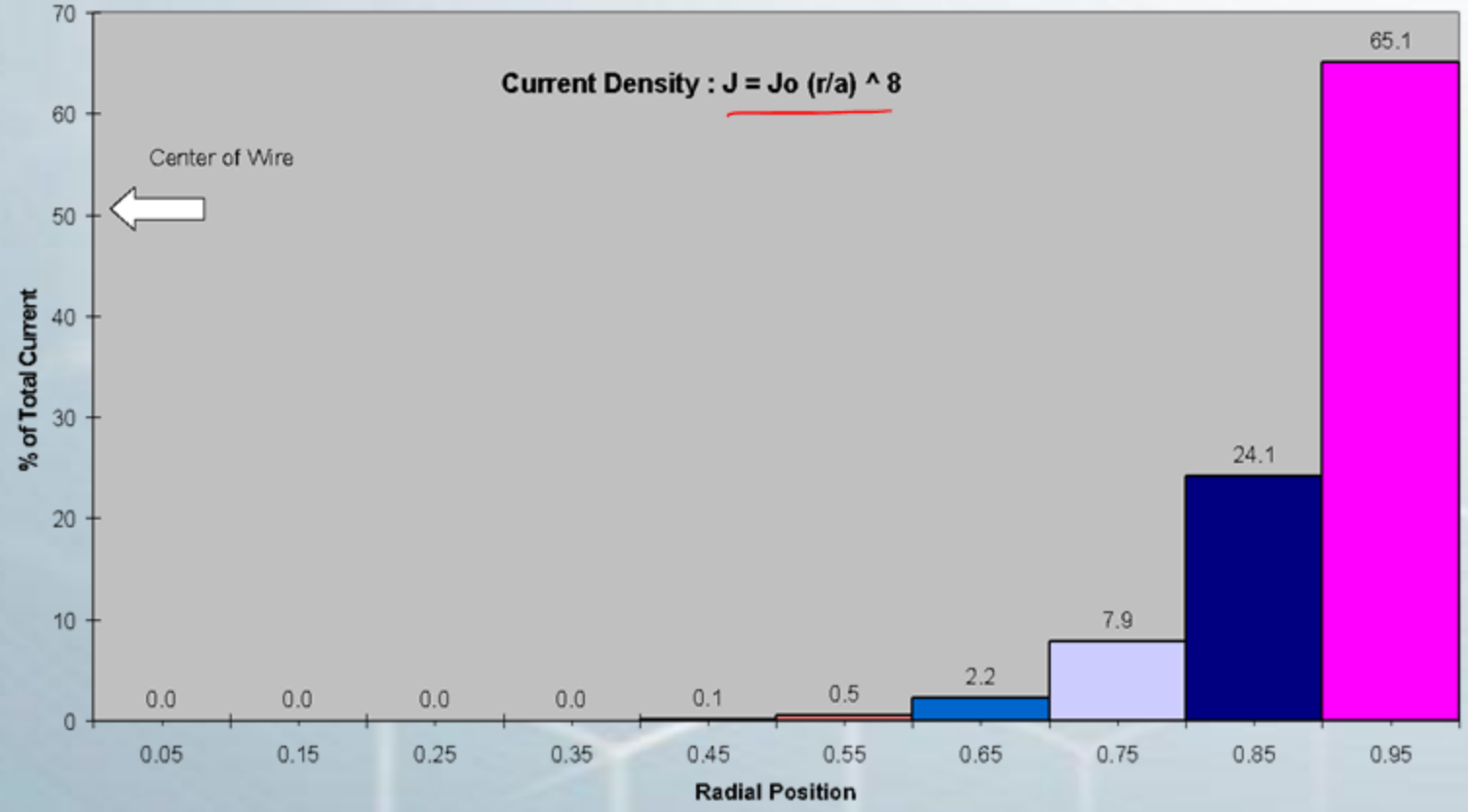


$$j = \frac{I}{Area} \cong \frac{I}{2 \cdot \pi \cdot a \cdot \delta}$$

But, $\delta \xrightarrow{limit} 0 \Rightarrow j \to \infty$

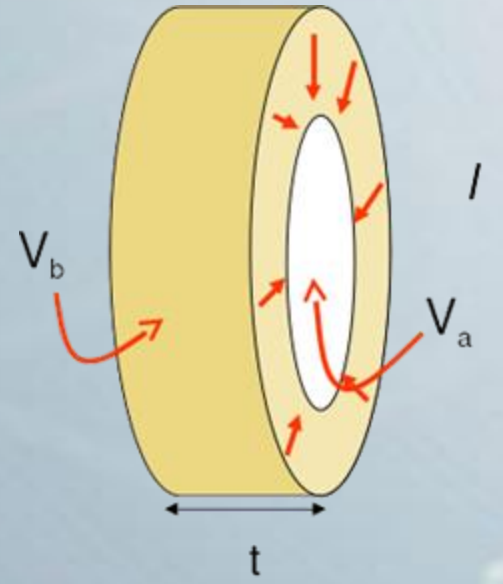
Current and Resistance

- Note that for this current distribution, there is much more current in the outer shell. This is a combination of the increased density and the larger area of each shell.



Current and Resistance

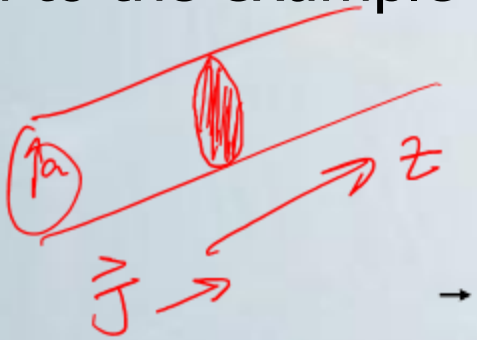
- $R = \frac{l}{\sigma A}$ is only valid when $\vec{J}$ and A are constant
- Therefore, in general, we must use the following equation:



$$R = \frac{V}{I} = \frac{-\int \vec{E} \cdot d\vec{l}}{\iint \vec{J} \cdot d\vec{s}} = \boxed{\frac{-\int \vec{E} \cdot d\vec{l}}{\sigma \iint \vec{E} \cdot d\vec{s}}}$$

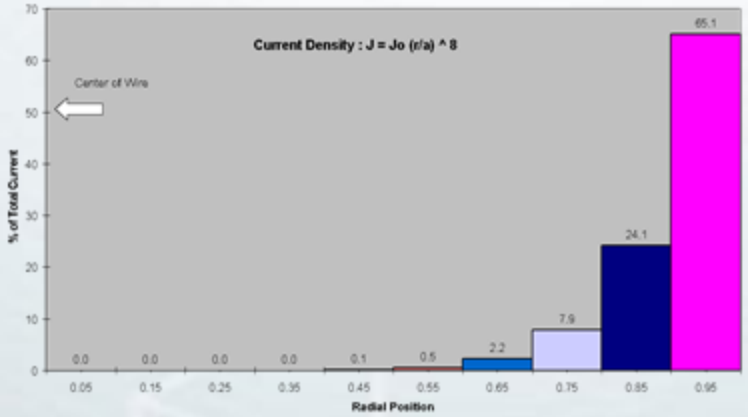
Current and Resistance

- Let's return to the example of a wire at higher frequencies:



a - Find I in terms of J_0 for $\vec{j} = j_0 \cdot \left(\frac{r}{a}\right)^8 \cdot \hat{a}_z$

Peak Density



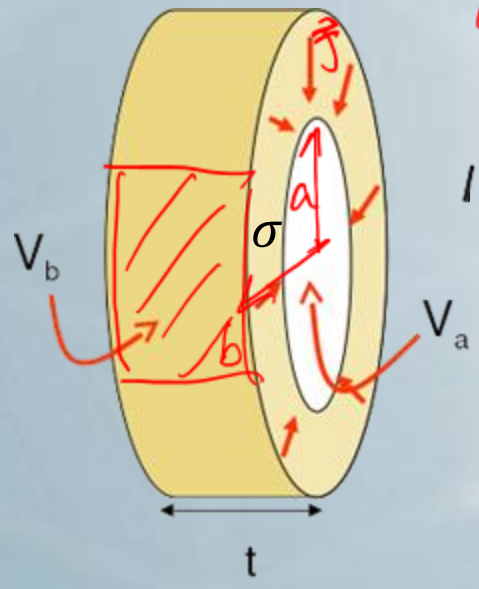
$$I = \iint \vec{J} \cdot d\vec{s} = \int_0^{2\pi} \int_0^{a} \left(j_0 \cdot \left(\frac{r}{a}\right)^8 \hat{z}\right) \cdot (r\,dr\,d\phi\,\hat{z}) = \frac{2\pi j_0}{a^8} \int_0^{} r^9\,dr = \frac{2\pi j_0}{a^8}\left(\frac{a^{10}}{10}\right)$$

$$= \frac{\pi j_0 a^2}{5}$$

Do Lecture 14 Exercise in groups of up to 4.

Current and Resistance

- Example: calculate the resistance of the cylindrical structure below.



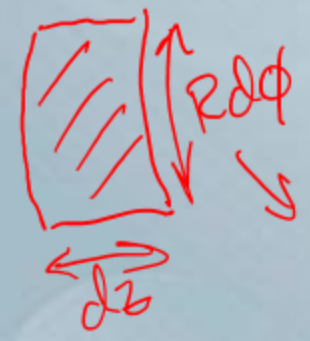
assume $V_b > V_a$

I is the same at the inner and outer part of disk

$$R = \frac{V}{I} = \frac{V_b - V_a}{\int \vec{J} \cdot d\vec{S}}$$

$$\vec{E} = \frac{V_b - V_a}{\ln(b/a)} \frac{1}{r} \hat{r} \Rightarrow I = \int_0^t \int_0^{2\pi} \left(\sigma \frac{V_b - V_a}{\ln(b/a)} \frac{1}{a} \hat{r}\right) \cdot (a\, d\phi\, dz)$$

$$= 2\pi t \left(\frac{V_b - V_a}{\ln(b/a)}\right)$$



what is R?

$$R = V/I = \frac{\ln(b/a)}{2\pi t}$$

Current and Resistance

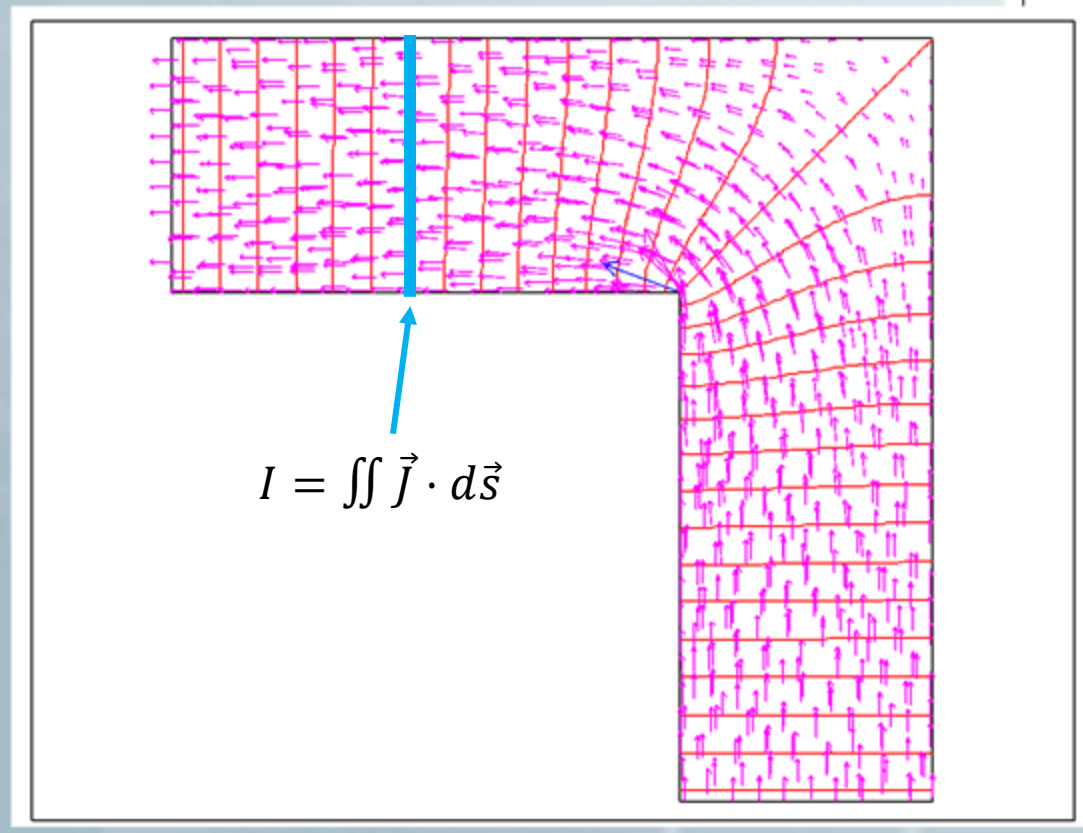
Numerical Methods for Current

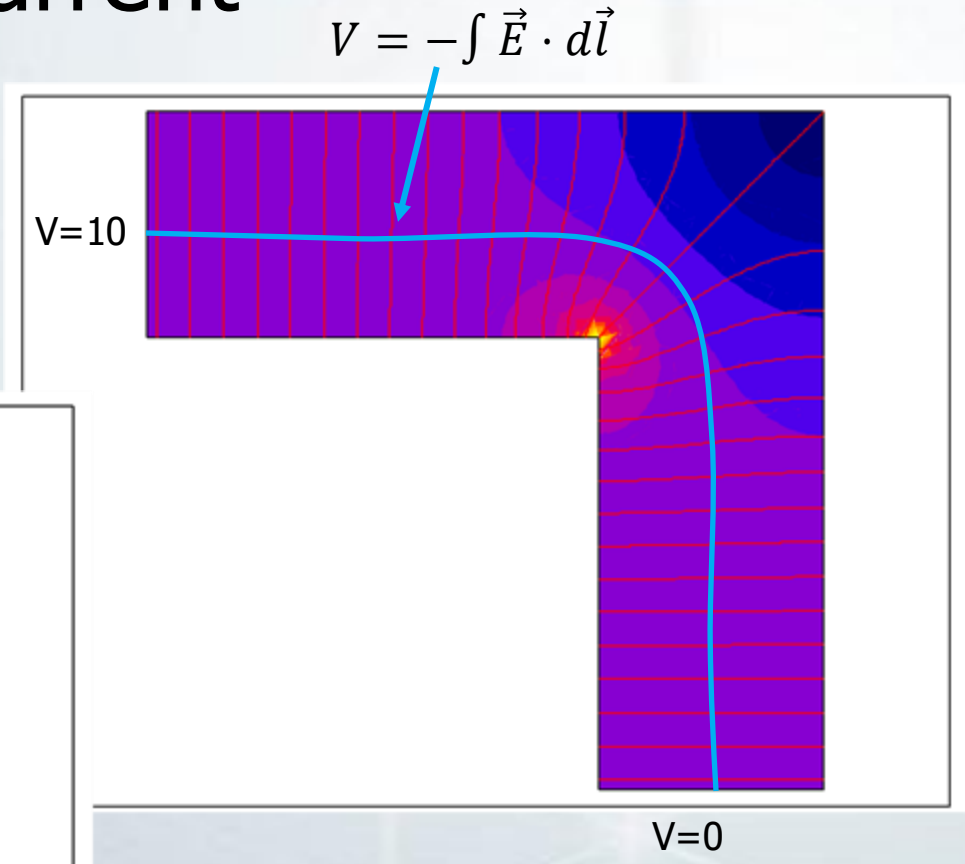
- Since current density is proportional to (and parallel to) the electric field, we can represent $\vec{J}$ in finite element and finite difference analysis similarly to how we would represent $\vec{E}$
  - In finite difference, just multiply $\vec{E}$ by the $\boldsymbol{\sigma}$ of the cell to get $\vec{J}$

- To calculate resistance, note that we are addressing the flow of electrons (current) **across** equi-potentials (E field perpendicular to equipotential lines)
  - In finite difference, calculate the total current I (from $\vec{J}$) and divide by the potential difference V to get R for a geometry

Numerical Methods for Current

$$R = \frac{V}{I} = \frac{-\int \vec{E} \cdot d\vec{l}}{\iint \vec{J} \cdot d\vec{s}}$$





Current vector field represented by pink arrows; red lines are equi-potentials

Why is vector field pointing from low to high voltage?

Wrap Up

- Homework 6 due Wednesday
- Extra office hours instead of studio session on Wednesday
- Exam 2 next Monday